Universidad de Alcalá

Estudio: **MÁSTER DE FORMACIÓN PERMANENTE EN GENERATIVE AI AND AI AGENTS GOVERNANCE**

Código Plan de Estudios: **FF17**

Año Académico: **2025-2026**

ESTRUCTURA GENERAL DEL PLAN DE ESTUDIOS:							
CURSO	Obligatorios		Optativos		Prácticas Externas	TFM/Memoria/ Proyecto	Créditos Totales
	Créditos	Nº Asignaturas	Créditos	Nº Asignaturas	Créditos	Créditos	
1º	48	10				12	60
2º							
ECTS TOTALES	**48**	**10**				**12**	**60**

PROGRAMA TEMÁTICO:

ASIGNATURAS OBLIGATORIAS

Código Asignatura	Curso	Denominación	Carácter OB/OP	Créditos
708003	1	FOUNDATIONS OF GENERATIVE AI AND LARGE LANGUAGE MODELS	OB	9
708004	1	AI AND CYBERSECURITY IN THE AGE OF GENERATIVE MODELS	OB	6
708005	1	AI STRATEGY AND GOVERNANCE IN THE ENTERPRISE	OB	4,5
708006	1	AI FOR BUSINESS PROCESS AUTOMATION AND KNOWLEDGE WORK	OB	4,5
708007	1	AUTONOMOUS AGENTS AND MULTI-AGENT SYSTEMS	OB	4,5
708008	1	HUMAN-CENTRED DESIGN AND INTERACTION WITH GENERATIVE AGENTS	OB	4,5
708009	1	ETHICS, EXPLAINABILITY AND RESPONSIBLE GENERATIVE AI	OB	4,5
708010	1	LEGAL FRAMEWORKS AND INTELECTUAL PROPERTY IN THE AGE OF GENERATIVE AI	OB	4,5
708011	1	EVALUATION AND ALIGNMENT OF GENERATIVE AI SYSTEMS	OB	3
708012	1	MULTIMODAL GENERATIVE AI AND DIGITAL MEDIA GOVERNANCE	OB	3

TRABAJO FIN DEMÁSTER/MEMORIA /PROYECTO

Código Asignatura	Curso	Denominación	Carácter OB/OP	Créditos
708013	1	MASTER’S THESIS	OB	12

Carácter: OB - Obligatoria; OP – Optativa

Universidad de Alcalá

DOCENTE

Año académico	2025-2026	
Estudio	Máster de Formación Permanente en Generative AI and AI Agents Governance	
Nombre de la asignatura	FOUNDATIONS OF GENERATIVE AI AND LARGE LANGUAGE MODELS	
Carácter (Oblig./Opt./Prácticas/TFM)	Obligatoria	
Créditos (1 ECTS=25 horas)	9	
Modalidad (elegir una opción)		Presencial (más del 80% de las sesiones son presenciales)
		Híbrida (sesiones on-line entre el 40% y 60%, resto presencial)
	X	Virtual (al menos el 80% de las sesiones son on-line o virtuales)
Profesor/a responsable	Dr. José Ignacio Olmeda Martos	
Idioma en el que se imparte	Inglés	

PROFESORES IMPLICADOS EN LA DOCENCIA

Dr. José Ignacio Olmeda Martos

DISTRIBUCIÓN DE HORAS

Número de horas presenciales/on-line asistencia profesor/a	63
Número de horas de trabajo personal del estudiante	162
Total horas	225

CONTENIDOS (Temario)

Introduction to Deep Learning and transformer architectures, tokenization, embeddings, and autoregressive modeling. Core training paradigms: pre-training, fine-tuning, reinforcement learning from human feedback (RLHF), and retrieval-augmented generation (RAG). Structured prompt design and input/output formatting. Evaluation of open-source models (e.g., LLaMA) vs. commercial APIs Practical applications in customer service, marketing insights, and internal documentation. IOther issues: memory management, and vector-based retrieval.

RESULTADOS DE APRENDIZAJE (indicar un mínimo de tres y máximo de cinco)

- Understand the architecture and functioning of transformer-based language models
- Understand the foundational training paradigms: pre-training, fine-tuning, RLHF, and RAG
- Understand the role of embeddings, tokenization, and autoregressive generation in LLMs
- Understand the differences and trade-offs between open-source and commercial LLMs
- Understand the practical applications of generative models across key business functions

SISTEMA DE EVALUACIÓN
The program assesses learning through Continuous Assessment Tests (PEC) with both theoretical and practical components. These tests are scheduled within each course and include specific evaluation criteria.

SEGUIMIENTO DEL PROGRESO DEL ESTUDIANTE
Continuous assessment.

BIBLIOGRAFÍA
Bommasani, R., et al. (2022). *On the Opportunities and Risks of Foundation Models*. Stanford Center for Research on Foundation Models (CRFM), 214 págs.

Universidad de Alcalá

GUÍA DOCENTE

Año académico	2025-2026
Estudio	Máster de Formación Permanente en Generative AI and AI Agents Governance
Nombre de la asignatura	AI AND CYBERSECURITY IN THE AGE OF GENERATIVE MODELS
Carácter (Oblig./Opt./Prácticas/TFM)	Obligatoria
Créditos (1 ECTS=25 horas)	6

Modalidad (elegir una opción)		Presencial (más del 80% de las sesiones son presenciales)
		Híbrida (sesiones on-line entre el 40% y 60%, resto presencial)
	X	Virtual (al menos el 80% de las sesiones son on-line o virtuales)

Profesor/a responsable	Dr. José Ignacio Olmeda Martos
Idioma en el que se imparte	Inglés

PROFESORES IMPLICADOS EN LA DOCENCIA

Dr. José Ignacio Olmeda Martos

DISTRIBUCIÓN DE HORAS

Número de horas presenciales/on-line asistencia profesor/a	42
Número de horas de trabajo personal del estudiante	108
Total horas	150

CONTENIDOS (Temario)

This course explores the emerging cybersecurity risks associated with the deployment of generative AI and large language models (LLMs) in enterprise settings. Students will examine attack vectors such as prompt injection, model inversion, data leakage, and adversarial prompting. The course introduces red-teaming methodologies and security design principles such as zero-trust architecture, sandboxing, and access control. Learners will evaluate legal, reputational, and operational risks and design incident response plans that ensure resilience in AI-enabled organizations.

RESULTADOS DE APRENDIZAJE (indicar un mínimo de tres y máximo de cinco)

- Understand the main cybersecurity threats posed by generative AI models
- Identify common attack vectors including prompt injection and model inversion
- Design basic enterprise-level defenses using zero-trust and sandboxing principles
- Assess legal and reputational risks related to AI misuse or data leakage
- Formulate incident response strategies for LLM-driven business environments

SISTEMA DE EVALUACIÓN
The program assesses learning through Continuous Assessment Tests (PEC) with both theoretical and practical components. These tests are scheduled within each course and include specific evaluation criteria.

SEGUIMIENTO DEL PROGRESO DEL ESTUDIANTE
Continuous assessment

BIBLIOGRAFÍA
Vaidya, J., Gabbouj, M. and Li, J. (eds.) (2024) *Artificial Intelligence Security and Privacy*. Lecture Notes in Computer Science, Vol. 14509.

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GUÍA DOCENTE

Año académico	2025-2026
Estudio	Máster de Formación Permanente en Generative AI and AI Agents Governance
Nombre de la asignatura	AI STRATEGY AND GOVERNANCE IN THE ENTERPRISE
Carácter (Oblig./Opt./Prácticas/TFM)	Obligatoria
Créditos (1 ECTS=25 horas)	4.5

<table>
<tr><td rowspan="3">Modalidad (elegir una opción)</td><td></td><td>Presencial (más del 80% de las sesiones son presenciales)</td></tr>
<tr><td></td><td>Híbrida (sesiones on-line entre el 40% y 60%, resto presencial)</td></tr>
<tr><td>X</td><td>Virtual (al menos el 80% de las sesiones son on-line o virtuales)</td></tr>
<tr><td>Profesor/a responsable</td><td colspan="2">Dr. José Ignacio Olmeda Martos</td></tr>
<tr><td>Idioma en el que se imparte</td><td colspan="2">Inglés</td></tr>
</table>

PROFESORES IMPLICADOS EN LA DOCENCIA

Prof. Juan Avilés

DISTRIBUCIÓN DE HORAS

Número de horas presenciales/on-line asistencia profesor/a	31.5
Número de horas de trabajo personal del estudiante	81
Total horas	112.5

CONTENIDOS (Temario)

This course examines how organizations can strategically integrate generative AI and autonomous agents into their core business functions. It focuses on aligning AI initiatives with corporate goals, competitive positioning, innovation cycles, and value creation. Students explore enterprise governance models, including decision-making frameworks, AI steering committees, data ownership roles, and cross-functional collaboration. Topics include portfolio-level AI investment appraisal, operational readiness, internal risk escalation procedures, and change management in environments undergoing AI-driven transformation. Emphasis is placed on executive leadership, organizational agility, and aligning technical capabilities with strategic intent.

RESULTADOS DE APRENDIZAJE (indicar un mínimo de tres y máximo de cinco)

- Understand how generative AI creates business value across industries
- Design AI strategies aligned with organizational vision and KPIs
- Develop internal governance mechanisms for AI oversight and coordination
- Evaluate enterprise readiness and resource allocation for AI transformation
- Lead cross-functional initiatives for AI-driven innovation and change

SISTEMA DE EVALUACIÓN
The program assesses learning through Continuous Assessment Tests (PEC) with both theoretical and practical components. These tests are scheduled within each course and include specific evaluation criteria.

SEGUIMIENTO DEL PROGRESO DEL ESTUDIANTE
Continuous assessment.

BIBLIOGRAFÍA
McKinsey & Company. (2024). The State of AI: How Organizations Are Rewiring to Capture Value. McKinsey Global Institute. Available at: https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai-2024

Universidad de Alcalá

GUÍA DOCENTE

Año académico	2025-2026
Estudio	Máster de Formación Permanente en Generative AI and AI Agents Governance
Nombre de la asignatura	AI FOR BUSINESS PROCESS AUTOMATION AND KNOWLEDGE WORK
Carácter (Oblig./Opt./Prácticas/TFM)	Obligatoria
Créditos (1 ECTS=25 horas)	4.5

Modalidad (elegir una opción)		Presencial (más del 80% de las sesiones son presenciales)
		Híbrida (sesiones on-line entre el 40% y 60%, resto presencial)
	X	Virtual (al menos el 80% de las sesiones son on-line o virtuales)

Profesor/a responsable	Dr. José Ignacio Olmeda Martos
Idioma en el que se imparte	Inglés

PROFESORES IMPLICADOS EN LA DOCENCIA

Prof. Juan Avilés

DISTRIBUCIÓN DE HORAS

Número de horas presenciales/on-line asistencia profesor/a	31.5
Número de horas de trabajo personal del estudiante	81
Total horas	112.5

CONTENIDOS (Temario)

This course explores how generative AI and intelligent automation are transforming enterprise workflows, particularly in areas such as customer service, HR, finance, legal, compliance, and knowledge management. Students will examine the design of AI agents, orchestration frameworks, and retrieval-augmented generation (RAG) systems to automate document-heavy and decision-intensive processes. Emphasis is placed on identifying high-impact use cases, integrating AI into existing systems, and managing organizational change to maximize value creation and minimize disruption.

RESULTADOS DE APRENDIZAJE (indicar un mínimo de tres y máximo de cinco)

- Understand the potential of generative AI in automating business processes and enhancing knowledge work.
- Identify and evaluate high-impact use cases for AI-driven automation across various enterprise functions.
- Design and implement AI workflows.
- Assess the organizational readiness for AI adoption and develop strategies for change management.
- Measure the impact of AI automation initiatives on productivity, cost savings, and employee satisfaction.

SISTEMA DE EVALUACIÓN
The program assesses learning through Continuous Assessment Tests (PEC) with both theoretical and practical components. These tests are scheduled within each course and include specific evaluation criteria.

SEGUIMIENTO DEL PROGRESO DEL ESTUDIANTE
Continuous assessment

BIBLIOGRAFÍA
Wiesmüller, S. (2023). The Relational Governance of Artificial Intelligence. Springer.

GUÍA DOCENTE

Año académico	2025-2026	
Estudio	Máster de Formación Permanente en Generative AI and AI Agents Governance	
Nombre de la asignatura	AUTONOMOUS AGENTS AND MULTI-AGENT SYSTEMS	
Carácter (Oblig./Opt./Prácticas/TFM)	Obligatoria	
Créditos (1 ECTS=25 horas)	4.5	
Modalidad (elegir una opción)		Presencial (más del 80% de las sesiones son presenciales)
		Híbrida (sesiones on-line entre el 40% y 60%, resto presencial)
	X	Virtual (al menos el 80% de las sesiones son on-line o virtuales)
Profesor/a responsable	Dr. José Ignacio Olmeda Martos	
Idioma en el que se imparte	Inglés	

PROFESORES IMPLICADOS EN LA DOCENCIA

Prof. Daniel Martínez

DISTRIBUCIÓN DE HORAS

Número de horas presenciales/on-line asistencia profesor/a	31.5
Número de horas de trabajo personal del estudiante	81
Total horas	112.5

CONTENIDOS (Temario)

This course introduces students to the design, coordination, and deployment of autonomous and multi-agent systems in enterprise contexts. Topics include reactive, deliberative, and hybrid agent architectures; agent communication protocols; and task coordination strategies. Students explore how intelligent agents can be used for decision support, dynamic resource allocation, customer interaction, and business automation. The course includes hands-on work with frameworks for building AI agents using modern toolchains such as LangChain, OpenAI APIs, and ReAct. Students will learn how to create task-driven agents that interact with external data sources, tools, and memory to carry out business-relevant objectives with minimal human supervision.

RESULTADOS DE APRENDIZAJE (indicar un mínimo de tres y máximo de cinco)

- Understand the foundational principles of autonomous and multi-agent systems
- Classify and design agents using reactive, deliberative, or hybrid models
- Apply coordination strategies in multi-agent environments
- Build functional AI agents using tools such as LangChain and ReAct
- Evaluate the capabilities and limits of agent-based systems in enterprise workflows

SISTEMA DE EVALUACIÓN
The program assesses learning through Continuous Assessment Tests (PEC) with both theoretical and practical components. These tests are scheduled within each course and include specific evaluation criteria.

SEGUIMIENTO DEL PROGRESO DEL ESTUDIANTE
Continuous assessment

BIBLIOGRAFÍA
Lanham, M. (2023) *AI Agents in Action: Building Intelligent Agents Using ChatGPT, ReAct, LangChain, and More*. O'Reilly Media.

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Año académico	2025-2026
Estudio	Máster de Formación Permanente en Generative AI and AI Agents Governance
Nombre de la asignatura	HUMAN-CENTRED DESIGN AND INTERACTION WITH GENERATIVE AGENTS
Carácter (Oblig./Opt./Prácticas/TFM)	Obligatoria
Créditos (1 ECTS=25 horas)	4.5

Modalidad (elegir una opción)		Presencial (más del 80% de las sesiones son presenciales)
		Híbrida (sesiones on-line entre el 40% y 60%, resto presencial)
	X	Virtual (al menos el 80% de las sesiones son on-line o virtuales)

Profesor/a responsable	Dr. José Ignacio Olmeda Martos
Idioma en el que se imparte	Inglés

PROFESORES IMPLICADOS EN LA DOCENCIA

Dr. Luis Usero

DISTRIBUCIÓN DE HORAS

Número de horas presenciales/on-line asistencia profesor/a	31.5
Número de horas de trabajo personal del estudiante	81
Total horas	112.5

CONTENIDOS (Temario)

This course explores the intersection of user experience (UX) design and generative AI, focusing on the creation of intuitive, transparent, and value-aligned interactions between humans and AI agents. Students examine the principles of human-centered design, conversational UX, trust calibration, feedback loops, and user-facing agent interfaces in contexts such as customer service, onboarding, decision support, and coaching. Rapid advances in AI are enabling increasingly sophisticated agent systems that transform organizational performance by enhancing efficiency, effectiveness, and service quality. This course draws from conceptual and empirical studies to critically assess how human-centered AI can be implemented across industries such as finance, healthcare, retail, and aerospace, empowering students to define and plan responsible workplace applications.

RESULTADOS DE APRENDIZAJE (indicar un mínimo de tres y máximo de cinco)

- Understand the foundations of human-centered design in AI-driven systems
- Design usable, ethical, and trustworthy interfaces with generative agents
- Prototype and evaluate agent-user interactions using UX methods and trust metrics
- Analyze cross-industry use cases of human-AI collaboration
- Plan the deployment of human-centered AI agents within enterprise environments

SISTEMA DE EVALUACIÓN
The program assesses learning through Continuous Assessment Tests (PEC) with both theoretical and practical components. These tests are scheduled within each course and include specific evaluation criteria.

SEGUIMIENTO DEL PROGRESO DEL ESTUDIANTE
Continuous assessment.

BIBLIOGRAFÍA
Constantinos K. Coursaris, Joerg Beringer, Pierre-Majorique Léger, Burak Öz Eds. (2025) The Design of Human-Centered Artificial Intelligence for the Workplace. Springer.

Universidad de Alcalá

GUÍA DOCENTE

Año académico	2025-2026
Estudio	Máster de Formación Permanente en Generative AI and AI Agents Governance
Nombre de la asignatura	ETHICS, EXPLAINABILITY AND RESPONSIBLE GENERATIVE AI
Carácter (Oblig./Opt./Prácticas/TFM)	Obligatoria
Créditos (1 ECTS=25 horas)	4.5

Modalidad (elegir una opción)		Presencial (más del 80% de las sesiones son presenciales)
		Híbrida (sesiones on-line entre el 40% y 60%, resto presencial)
	X	Virtual (al menos el 80% de las sesiones son on-line o virtuales)

Profesor/a responsable	Dr. José Ignacio Olmeda Martos
Idioma en el que se imparte	Inglés

PROFESORES IMPLICADOS EN LA DOCENCIA

Prof. Carlos García

DISTRIBUCIÓN DE HORAS

Número de horas presenciales/on-line asistencia profesor/a	31.5
Número de horas de trabajo personal del estudiante	81
Total horas	112.5

CONTENIDOS (Temario)

This course explores the ethical foundations and applied principles guiding the responsible development and deployment of generative AI systems. Topics include fairness, accountability, transparency, non-discrimination, and privacy, as well as the challenges of aligning opaque AI models with human values. Special emphasis is placed on explainability as model auditing strategies that address bias, robustness, and compliance with ethical standards. The course further examines the implications of generative AI in high-stakes domains such as recruitment, credit scoring, legal decision-making, and healthcare. Through case studies and simulations, students will critically engage with ethical dilemmas and design governance frameworks that balance innovation with social responsibility and reputational risk management.

RESULTADOS DE APRENDIZAJE (indicar un mínimo de tres y máximo de cinco)

- Understand the foundational ethical principles guiding responsible AI development.
- Analyze the importance of explainability and interpretability in AI systems.
- Identify potential biases and develop strategies to mitigate them in AI models.
- Evaluate the societal and organizational impacts of generative AI technologies.
- Develop frameworks for the ethical governance and deployment of AI systems

SISTEMA DE EVALUACIÓN
The program assesses learning through Continuous Assessment Tests (PEC) with both theoretical and practical components. These tests are scheduled within each course and include specific evaluation criteria.

SEGUIMIENTO DEL PROGRESO DEL ESTUDIANTE
Continuous assessment

BIBLIOGRAFÍA
Smith, J. J., Deng, W. H., Smith, W. H., Sap, M., DeCario, N., & Dodge, J. (2024). *The Generative AI Ethics Playbook*. arXiv preprint arXiv:2501.1038, 63 págs.

GUÍA DOCENTE

Año académico	2025-2026	
Estudio	Máster de Formación Permanente en Generative AI and AI Agents Governance	
Nombre de la asignatura	LEGAL FRAMEWORKS AND INTELLECTUAL PROPERTY IN THE AGE OF GENERATIVE AI	
Carácter (Oblig./Opt./Prácticas/TFM)	Obligatoria	
Créditos (1 ECTS=25 horas)	4.5	
Modalidad (elegir una opción)		Presencial (más del 80% de las sesiones son presenciales)
		Híbrida (sesiones on-line entre el 40% y 60%, resto presencial)
	X	Virtual (al menos el 80% de las sesiones son on-line o virtuales)
Profesor/a responsable	Dr. José Ignacio Olmeda Martos	
Idioma en el que se imparte	Inglés	

PROFESORES IMPLICADOS EN LA DOCENCIA

Prof. Carlos García

DISTRIBUCIÓN DE HORAS

Número de horas presenciales/on-line asistencia profesor/a	31.5
Número de horas de trabajo personal del estudiante	81
Total horas	112.5

CONTENIDOS (Temario)

This course explores the evolving legal landscape surrounding generative AI, with a particular focus on intellectual property (IP), data governance, and liability. Students examine how legal systems address ownership of AI-generated content, protection of training data, and downstream responsibility for AI-driven outputs. Core topics include copyright, database rights, trade secrets, and the legal treatment of synthetic media and derivative works. A strong comparative dimension is introduced through the analysis of key international regulatory approaches. Students assess the emerging EU AI Act alongside U.S. sector-specific regulation and China's model of ex-ante content control and algorithm registration. Case studies from the creative industries, enterprise AI, and open-source development are used to contextualize legal risks and innovation strategies. The course equips students to navigate cross-border legal challenges and to draft baseline clauses governing IP ownership, data licensing, and liability in AI development and deployment.

RESULTADOS DE APRENDIZAJE (indicar un mínimo de tres y máximo de cinco)

- Understand how different jurisdictions approach intellectual property rights in the context of AI-generated content
- Compare major legal and regulatory frameworks for AI governance in the EU, U.S., and China

- Analyze the legal implications of using third-party training data and open-source models
- Identify privacy, transparency, and accountability requirements for generative AI systems
- Understand and evaluate contractual clauses related to AI output ownership, liability, and data licensing

SISTEMA DE EVALUACIÓN

The program assesses learning through Continuous Assessment Tests (PEC) with both theoretical and practical components. These tests are scheduled within each course and include specific evaluation criteria.

SEGUIMIENTO DEL PROGRESO DEL ESTUDIANTE

Continuous assessment

BIBLIOGRAFÍA

Cwik, C. H., Suarez, C. A., & Thomson, L. L. (eds.) (2024). Artificial Intelligence: Legal Issues, Policy, and Practical Strategies. Chicago: American Bar Association.

Universidad de Alcalá

GUÍA DOCENTE

Año académico	2025-2026
Estudio	Máster de Formación Permanente en Generative AI and AI Agents Governance
Nombre de la asignatura	EVALUATION AND ALIGNMENT OF GENERATIVE AI SYSTEMS
Carácter (Oblig./Opt./Prácticas/TFM)	Obligatoria
Créditos (1 ECTS=25 horas)	3

Modalidad (elegir una opción)		Presencial (más del 80% de las sesiones son presenciales)
		Híbrida (sesiones on-line entre el 40% y 60%, resto presencial)
	X	Virtual (al menos el 80% de las sesiones son on-line o virtuales)
Profesor/a responsable	Dr. José Ignacio Olmeda Martos	
Idioma en el que se imparte	Inglés	

PROFESORES IMPLICADOS EN LA DOCENCIA

Prof. Daniel Martínez

DISTRIBUCIÓN DE HORAS

Número de horas presenciales/on-line asistencia profesor/a	21
Número de horas de trabajo personal del estudiante	54
Total horas	75

CONTENIDOS (Temario)

This course explores how organizations can evaluate and align generative AI systems to ensure they perform reliably, responsibly, and in accordance with human goals. Rather than focusing on low-level technical metrics, the course introduces practical frameworks for assessing the usefulness, safety, fairness, and reputational impact of AI systems across sectors. Key themes include human-in-the-loop evaluation, alignment with organizational values, and the development of performance benchmarks that go beyond accuracy to include trust, transparency, and social impact. Students will learn how to compare AI evaluation strategies adopted by leading firms, standard-setting bodies, and academic institutions. The course emphasizes risk-aware design and the governance of model behavior through clear criteria, oversight structures, and continuous monitoring practices.

RESULTADOS DE APRENDIZAJE (indicar un mínimo de tres y máximo de cinco)

- Understand the purpose and importance of evaluation and alignment in generative AI systems
- Identify practical evaluation strategies used in industry and public-sector AI deployments
- Analyze alignment challenges related to bias, misuse, and value conflicts
- Design evaluation criteria and governance mechanisms aligned with business and social priorities
- Interpret the implications of emerging standards for trustworthy and auditable AI

SISTEMA DE EVALUACIÓN
The program assesses learning through Continuous Assessment Tests (PEC) with both theoretical and practical components. These tests are scheduled within each course and include specific evaluation criteria.

SEGUIMIENTO DEL PROGRESO DEL ESTUDIANTE
Continuous assessment

BIBLIOGRAFÍA
Weidinger, L. et al. (2025) Toward an Evaluation Science for Generative AI Systems. arXiv preprint arXiv:2503.05336.

GUÍA DOCENTE

Año académico	2025-2026	
Estudio	Máster de Formación Permanente en Generative AI and AI Agents Governance	
Nombre de la asignatura	MULTIMODAL GENERATIVE AI AND DIGITAL MEDIA GOVERNANCE	
Carácter (Oblig./Opt./Prácticas/TFM)	Obligatoria	
Créditos (1 ECTS=25 horas)	3	
Modalidad (elegir una opción)		Presencial (más del 80% de las sesiones son presenciales)
		Híbrida (sesiones on-line entre el 40% y 60%, resto presencial)
	X	Virtual (al menos el 80% de las sesiones son on-line o virtuales)
Profesor/a responsable	Dr. José Ignacio Olmeda Martos	
Idioma en el que se imparte	Inglés	

PROFESORES IMPLICADOS EN LA DOCENCIA

Dr. Juan Pedro Llerena

DISTRIBUCIÓN DE HORAS

Número de horas presenciales/on-line asistencia profesor/a	21
Número de horas de trabajo personal del estudiante	54
Total horas	75

CONTENIDOS (Temario)

This course explores the intersection of multimodal generative AI technologies and the governance of digital media. Students will examine how AI systems that process and generate multiple data types—such as text, images, audio, and video—are transforming content creation, dissemination, and consumption. The course delves into the implications of these technologies for media authenticity, intellectual property rights, and the spread of misinformation. Through case studies and policy analyses, students will assess regulatory frameworks and industry standards aimed at ensuring ethical and responsible use of multimodal AI in digital media. The course emphasizes the importance of transparency, accountability, and inclusivity in the development and deployment of these technologies.

RESULTADOS DE APRENDIZAJE (indicar un mínimo de tres y máximo de cinco)

- Understand the capabilities and applications of multimodal generative AI in digital media contexts.
- Analyze the ethical, legal, and societal challenges posed by AI-generated multimedia content.
- Evaluate existing governance structures and propose improvements for overseeing multimodal AI technologies.
- Develop strategies for promoting transparency and accountability in AI-driven media systems.
- Critically assess the impact of multimodal AI on media authenticity and public trust.

SISTEMA DE EVALUACIÓN
The program assesses learning through Continuous Assessment Tests (PEC) with both theoretical and practical components. These tests are scheduled within each course and include specific evaluation criteria.

SEGUIMIENTO DEL PROGRESO DEL ESTUDIANTE
Continuous assessment

BIBLIOGRAFÍA
Singh, A. & Singh, K. K. (eds.) (2025). Multimodal Generative AI. Singapore: Springer Nature.

Universidad de Alcalá

GUÍA DOCENTE

Año académico	2024-2025
Estudio	Máster de Formación Permanente en Generative AI and AI Agents Governance
Nombre de la asignatura	MASTER'S THESIS
Carácter (Obligatoria/Optativa/Prácticas/TFM)	TFM
Créditos (1 ECTS=25 horas)	12

Modalidad (elegir una opción)		Presencial (más del 80% de las sesiones son presenciales)
		Híbrida (sesiones on-line entre el 40% y 60%, resto presencial)
	x	Virtual (al menos el 80% de las sesiones son on-line o virtuales)

Profesor/es responsable/s	Dr. José Ignacio Olmeda Martos
Idioma en el que se imparte	Inglés

PROFESORES IMPLICADOS EN LA DOCENCIA

Dr. José Ignacio Olmeda Martos

DISTRIBUCIÓN DE HORAS

Número de horas presenciales/on-line asistencia profesor/a	84
Número de horas de trabajo personal del estudiante	216
Total horas	300

CONTENIDOS (Temario)

Independent short research paper conducted by the student on one of the Master's topics. The paper must be presented and defended before a committee at the end of the Master's program.

RESULTADOS DE APRENDIZAJE (indicar un mínimo de tres y máximo de cinco)

- Be able to search for and critically evaluate sources relevant to the design and implementation of generative AI systems or frameworks
- Be able to propose models or methodologies to address theoretical or applied problems in the field of generative AI, agents, or AI governance
- Be able to structure, write, and present a research-based thesis clearly and coherently, following academic standards
- Be able to propose innovative and responsible solutions in areas such as enterprise automation, AI strategy, or human–AI interaction
- Be able to reflect on the societal, ethical, or regulatory implications of the proposed AI solution or model

SISTEMA DE EVALUACIÓN
Thesis defense before a committee

BIBLIOGRAFÍA
The bibliography will be determined by the specific subject matter of the thesis